Formal Definitions, Results, and Provenance for Categorical Fine Graining

CFT Anyons Formalisation Workspace

May 14, 2026

Abstract

This document is the self-contained mathematical record for the formalisation project. Every external mathematical input is tied to a local source file. Every result is labelled as proved, ready for adversarial verification, in progress, or conjectural. A result is not accepted merely because it appears in the handoff document; acceptance requires local source provenance, a proof-tree entry in the Adversarial Proof Framework, and where applicable three independent checks in Lean, Julia, and WolframScript.

Contents

1 Status and Ground Rules

Definition 1.1 (Acceptance gate)**.** *A statement is accepted only after all applicable items below have been recorded.*

1. *A local source string locator for every external definition, equation, or claim.*
2. *A proof-tree node in the Adversarial Proof Framework.*

3. *A Lean proof whenever the claim is within the current formal scope.*

4. *A Julia check for numerical or finite examples.*

5. *A WolframScript check for exact symbolic algebra when the claim is algebraic.*

6. *A fresh adversarial verifier subagent has accepted the proof-tree node.*

Warning 1.2 (Theorem targets are not theorems yet)**.** *Statements labelled "target" or "conjecture" are not claimed as proved. They are included so that the document contains the full mathematical surface area of the project.*

2 Local Source Provenance

Source identifier	**Local source and use**
`SRC-FIB-TREBST-LOCAL`	`references/FibonacciAnyonModels.pdf`; extracted text in `references/text/FibonacciAnyonModels.txt`. Used for Fibonacci fusion rules, quantum dimension, the standard Fibonacci F-matrix, and finite Fibonacci R- and braid-matrix data.
`SRC-FIB-TREBST-ARXIV`	`references/TrebstShortIntroductionFibonacciAnyons.pdf`; extracted text in `references/text/TrebstShortIntroductionFibonacciAnyons.txt` Used as a second local source for the golden-ratio quantum dimension.
`SRC-GOLDEN-CHAIN`	`references/GoldenChainFeiguinEtAl.pdf`; extracted text in `references/text/GoldenChainFeiguinEtAl.txt`. Used for the same Fibonacci F-matrix.
`SRC-KZ-FIB`	`references/IsingLikeFibonacciAnyonsKZ.pdf`; extracted text in `references/text/IsingLikeFibonacciAnyonsKZ.txt`. Used for the conformal weight $h_\tau = 2/5$. The extracted text loses the fraction slash in the table row, so the row was checked visually against the local PDF page: the spin-1 field has $h = 2/5$.
`SRC-TL-JONES`	`references/TemperleyLiebRootsJonesQuotient.pdf`; extracted text in `references/text/TemperleyLiebRootsJonesQuotient.txt`. Used for the Temperley–Lieb and Jones quotient background.
`SRC-ISING-CHAIN`	`references/PoilblancOneDimensionalItinerantInteractingAnyons.pdf`; extracted text in `references/text/PoilblancOneDimensionalItinerantInteractin` Used for Ising labels, fusion rules, and the nonchiral dimensions $\Delta_\sigma = 1/8$, $\Delta_\psi = 1$.
`SRC-STRING-NET`	`references/StringNetCondMat0510613.pdf`; extracted text in `references/text/StringNetCondMat0510613.txt`. Used for string-net fixed-point data: F-symbols and quantum dimensions.
`SRC-PENNEYS-UFC`	`references/text/PenneysUnitaryFusionCategories.md`. Used for unitary fusion category background, direct sums, orthogonal direct sums, and polar decomposition language.
`SRC-DEFECT-FOCK`	`references/GaugingDefectsQuantumSpinSystems.pdf`; extracted text in `references/text/GaugingDefectsQuantumSpinSystems.txt`. Used for fusion category definitions, Fock-space direct sums, and fusion-tree Hilbert spaces.

SRC-OAR-FERMIONS and SRC-OAR-WAVELETS	Local extracted texts in `references/text/`. Used for operator-algebraic renormalization and isometric wavelet refinement maps.

3 Categorical Background

Definition 3.1 (Fusion category, local working form)**.** *Over the complex numbers, a fusion category is treated as a rigid semisimple tensor category with finitely many isomorphism classes of simple objects and a simple tensor unit. The local sources are* `references/text/GaugingDefectsQuantumSpinSystems.txt:` *and* `:404`*.*

Definition 3.2 (Unitary fusion category, local working form)**.** *A unitary fusion category is a fusion category with a dagger operation on morphisms and a compatible unitary monoidal structure. Orthogonal direct sums have isometric inclusions whose adjoints are projections. The local sources are* `references/text/PenneysUnitaryFusionCategories.md:219`*,* `:319`*,* `:456`*, and* `:458`*.*

Definition 3.3 (Skeletal finite model)**.** *For Lean formalisation, the initial model consists of a finite type of simple labels, a distinguished unit label, and natural-number fusion multiplicities*

$$N_{ab}^{c} \in \mathbb{N}.$$

The local skeletal-data source states that one chooses a finite set of simple objects with a distinguished unit and finite-dimensional fusion spaces. The Lean file `Formalisation/Foundations/SkeletalFusion.lean` *formalises only the finite label/multiplicity skeleton with left and right unit laws:*

$$N_{\mathbf{1}a}^{c} = \delta_{a,c}, \qquad N_{a\mathbf{1}}^{c} = \delta_{a,c}.$$

It also proves that the total unit fusion multiplicity is 1 *on either side. AF node* `1.2.7` *has been accepted by verifier* `verifier-finite-skeletal-fusion-data-001`*. This does not construct a fusion category, associators, F-symbols, pentagon equations, duals, Hilbert morphism spaces, or categorical reconstruction.*

Definition 3.4 (Local occupation object)**.** *Let $\mathcal{C}$ be the finite skeletal model. Define*

$$Q := \bigoplus_{a \in \mathrm{Irr}(\mathcal{C})} a.$$

The vacuum summand represents an empty site. For N ordered sites, define

$$Q^{\otimes N} := \underbrace{Q \otimes \cdots \otimes Q}_{N \text{ factors}}.$$

Status: the categorical direct-sum object is still a target. The accepted finite coordinate skeleton is a finite label type with a distinguished vacuum label; the summand labels of Q are represented by that label type, and the summand labels of $Q^{\otimes N}$ are ordered configurations $\mathrm{Fin}(N) \to A$*. This is formalised as* `LocalOccupationModel` *and* `LocalOccupationTensorSummand` *in* `Formalisation/Foundations/ProjectDefini`

Definition 3.5 (Indefinite-particle Hilbert space)**.** *For a total charge $W \in \mathrm{Irr}(\mathcal{C})$, define*

$$\mathcal{H}_N^W := \mathrm{Hom}_{\mathcal{C}}(W, Q^{\otimes N}).$$

If $\mathcal{C}$ is only a fusion category, this is a finite-dimensional complex vector space. If $\mathcal{C}$ is unitary, it is treated as a Hilbert space after choosing the standard inner products on fusion-tree bases. Status: the categorical Hom-space definition remains a target. The accepted finite coordinate skeleton assumes a finite coordinate basis for every configuration-sector $\mathrm{Hom}(W, a_1 \otimes \cdots \otimes a_N)$*. The flattened coordinates of the resulting fixed sector are formalised as* `IndefiniteParticleSectorCoordinates`*, with component coordinates and pairing preservation inherited from the configuration-space coordinate expansion.*

Definition 3.6 (Particle number)**.** *For a configuration $\vec{a} = (a_1, \ldots, a_N)$, define its particle number by*

$$\#\{i : a_i \neq \mathbf{1}\}.$$

Thus $\mathcal{H}_N^W$ contains sectors with particle number from 0 to N. In the finite Lean model, an ordered configuration of length N is a function $\mathrm{Fin}(N) \to A$, where A is the finite label type.

4 General Theorem Targets

Theorem 4.1 (Finite ordered configuration combinatorics)**.** *Let A be a finite label set and let N be a natural number. The set of ordered configurations $\mathrm{Fin}(N) \to A$ has cardinality*

$$|A|^N.$$

If a vacuum label is chosen, the particle number of any configuration is at most N. The constant vacuum configuration has particle number 0, and a constant non-vacuum configuration has particle number N. For Fibonacci labels $\{\mathbf{1}, \tau\}$, the number of configurations is 2^N. These finite combinatorial statements are proved in `Formalisation/Foundations/Configurations.lean`*; Julia and WolframScript check Fibonacci configuration counts and particle sectors through $N = 8$. AF node* `1.2.1` *has been accepted by verifier* `verifier-config-001`*. This theorem does not prove the categorical Hom-space expansion below.*

Theorem 4.2 (Finite direct-sum coordinate model)**.** *Let I be a finite set and let K_i be a finite coordinate basis for the i-th component. The space of flattened complex coordinates*

$$\{x : \coprod_{i \in I} K_i \to \mathbb{C}\}$$

is complex-linearly equivalent to the dependent family of component coordinate spaces

$$\prod_{i \in I} \{x_i : K_i \to \mathbb{C}\}.$$

The equivalence sends x to the family $x_i(k) = x(i,k)$, and its inverse sends a family x_i to the flattened function $(i,k) \mapsto x_i(k)$. It also satisfies

$$\left| \coprod_{i \in I} K_i \right| = \sum_{i \in I} |K_i|$$

and preserves the standard finite coordinate pairing

$$\sum_{i \in I} \sum_{k \in K_i} \overline{x_i(k)}\, y_i(k) = \sum_{(i,k) \in \coprod_i K_i} \overline{x(i,k)}\, y(i,k).$$

This is proved in `Formalisation/Foundations/DirectSumCoordinates.lean`*; Julia and WolframScript check exact rational complex examples. AF node* `1.2.2` *has been accepted by verifier* `verifier-dirsum-coord-` *This result is only the finite coordinate skeleton obtained after bases have been chosen in the summands; it does not construct categorical biproducts, Q, tensor powers, Hom functors, dagger categories, or fusion-category basis provenance.*

Theorem 4.3 (Finite direct-sum projection equations)**.** *In the same finite coordinate model, write*

$$\iota_i : \{x_i : K_i \to \mathbb{C}\} \to \{x : \coprod_{j\in I} K_j \to \mathbb{C}\}$$

for the inclusion that places component coordinates into the i-th block, and write

$$\pi_i : \{x : \coprod_{j\in I} K_j \to \mathbb{C}\} \to \{x_i : K_i \to \mathbb{C}\}$$

for the projection onto the i-th block. Then

$$\pi_i\iota_i = \mathrm{id}, \qquad \pi_i\iota_j = 0 \quad (i \neq j), \qquad \sum_{i\in I} \iota_i\pi_i = \mathrm{id}\,.$$

This is the finite coordinate analogue of Penneys' $\pi_i \circ \iota_j = \delta_{i=j}\,\mathrm{id}$ and $\sum_i \iota_i \circ \pi_i = \mathrm{id}$ equations. It is proved in `Formalisation/Foundations/DirectSumCoordinates.lean`*; Julia and WolframScript check exact finite matrix examples. AF node* `1.2.3` *has been accepted by verifier* `verifier-dirsum-proj-001`*. This result does not construct categorical biproducts or the direct-sum universal property.*

Theorem 4.4 (Finite orthogonal direct-sum coordinates)**.** *In the finite coordinate model, equip every component and the flattened direct sum with the standard complex coordinate pairing. Then each component inclusion ι_i is isometric:*

$$\langle \iota_i x, \iota_i y\rangle = \langle x, y\rangle.$$

The coordinate range map $P_i = \iota_i\pi_i$ satisfies

$$P_i^2 = P_i, \qquad P_iP_j = 0 \quad (i \neq j).$$

Moreover, component projection is the coordinate adjoint of component inclusion:

$$\langle \iota_i x, y\rangle = \langle x, \pi_i y\rangle, \qquad \langle y, \iota_i x\rangle = \langle \pi_i y, x\rangle,$$

and therefore the coordinate range projection is self-adjoint for the pairing:

$$\langle P_i x, y\rangle = \langle x, P_i y\rangle.$$

This is the finite coordinate analogue of Penneys' statement that orthogonal direct-sum inclusions are isometries and the maps $\iota_i\iota_i^\dagger$ are mutually orthogonal projections. It is proved in `Formalisation/Foundations/Dire` *Julia and WolframScript check exact finite matrix and pairing examples. AF nodes* `1.2.4` *and* `1.2.8` *have been accepted by verifiers* `verifier-dirsum-orth-001` *and* `verifier-dirsum-adjoint-coord-001`*. This result does not construct categorical adjoints or orthogonal direct sums.*

Theorem 4.5 (Finite configuration-space coordinate expansion)**.** *Let A be a finite label set and let*

$$\operatorname{Config}(A, N) = \{\, c : \operatorname{Fin}(N) \to A \,\}$$

be the finite set of ordered configurations. Suppose that every configuration c has been assigned a finite coordinate basis B_c. The space of flattened complex coordinates

$$\{x : \coprod_{c \in \operatorname{Config}(A,N)} B_c \to \mathbb{C}\}$$

is complex-linearly equivalent to the dependent family of component coordinate spaces

$$\prod_{c \in \operatorname{Config}(A,N)} \{x_c : B_c \to \mathbb{C}\}.$$

The equivalence sends x to the family $x_c(b) = x(c, b)$. Its coordinate cardinality satisfies

$$\left| \coprod_{c \in \operatorname{Config}(A,N)} B_c \right| = \sum_{c \in \operatorname{Config}(A,N)} |B_c|.$$

If all sectors have the same finite basis B, then

$$\left| \coprod_{c \in \operatorname{Config}(A,N)} B \right| = |A|^N \, |B|.$$

The equivalence preserves the standard finite coordinate pairing:

$$\sum_c \sum_{b \in B_c} \overline{x_c(b)} \, y_c(b) = \sum_{(c,b)} \overline{x(c, b)} \, y(c, b).$$

For Fibonacci labels $A = \{\mathbf{1}, \tau\}$, the constant-sector cardinality is $2^N|B|$. Lean proves these statements in `Formalisation/Foundations/ConfigurationSpace.lean` using the accepted configuration and direct-sum coordinate results. Julia and WolframScript check exact Fibonacci $N = 3$ examples with dependent sector sizes and rational complex pairings. AF node `1.2.5` has been accepted by verifier `verifier-config-space-coord-001`. This theorem is only the finite, basis-dependent coordinate skeleton after sector bases have been supplied; it does not construct categorical Hom spaces, $Q^{\otimes N}$, tensor products, canonical Hom isomorphisms, categorical direct sums, or unitarity in a fusion category.

Theorem 4.6 (Finite coordinate skeleton of Q and $\mathcal{H}_N^W$)**.** *Let A be a finite label type with a distinguished vacuum label. In the finite coordinate model, the summands of Q are indexed by A, and the summands of $Q^{\otimes N}$ are indexed by functions*

$$\operatorname{Fin}(N) \to A.$$

Hence there are $|A|^N$ label summands in $Q^{\otimes N}$, and the constant vacuum configuration has particle number 0. After choosing a finite basis for each sector $\operatorname{Hom}(W, a_1 \otimes \cdots \otimes a_N)$, the fixed sector $\mathcal{H}_N^W$ has flattened coordinates over all configuration-sector basis vectors. These coordinates expand to component coordinates over configurations, and the standard finite coordinate pairing is preserved.

Lean proves these statements in `Formalisation/Foundations/ProjectDefinitions.lean`*. Julia and WolframScript check the Fibonacci two-label instance, the $Q^{\otimes 4}$ label count, vacuum and all-τ particle numbers, flattened coordinate dimension, unflattening, and pairing preservation. AF node* `1.2.6` *has been accepted by verifier* `verifier-project-q-hilb-coord-001`*. This is only a finite, basis-dependent coordinate skeleton; it does not construct categorical direct sums, tensor powers, Hom functors, canonical categorical isomorphisms, or fusion-category Hilbert-space semantics.*

Theorem 4.7 (Finite truncated distinguishable-Fock coordinates)**.** *Let B be a finite local coordinate basis and let M be a nonnegative integer. The finite truncation of distinguishable Fock coordinates through particle number M has basis*

$$\coprod_{n=0}^{M} \{\, w : \mathrm{Fin}(n) \to B \,\}.$$

Equivalently, a basis vector is a particle number $n \le M$ together with an ordered word of length n in the local basis. Flattened coordinates on this coproduct are complex-linearly equivalent to the family of fixed-particle number coordinate vectors. The basis cardinality is

$$\left| \coprod_{n=0}^{M} \{w : \mathrm{Fin}(n) \to B\} \right| = \sum_{n=0}^{M} |B|^n.$$

The zero-particle sector has one basis vector, and the one-particle sector has $|B|$ basis vectors. The flattening by particle number preserves the standard finite coordinate pairing. For the Fibonacci two-label local basis, the cardinality is $\sum_{n=0}^{M} 2^n$.

This is the finite coordinate truncation of the locally sourced distinguishable-Fock construction as a direct sum over particle number, with a one-dimensional zero-particle vacuum sector. Lean proves the finite statements in `Formalisation/Foundations/FockSpace.lean`*. Julia and WolframScript check exact finite counts and rational complex pairings. AF node* `1.2.9` *has been accepted by verifier* `verifier-truncated-fock-coord-001`*. This result does not construct an infinite direct sum, a Hilbert-space completion, tensor-product Hilbert spaces, exchange-symmetry quotients, categorical Hom spaces, or physical Fock-space semantics.*

Theorem 4.8 (Configuration-space expansion target)**.** *There should be a canonical isomorphism*

$$\mathcal{H}_N^W \cong \bigoplus_{(a_1,\ldots,a_N)} \mathrm{Hom}_{\mathcal{C}}(W, a_1 \otimes \cdots \otimes a_N).$$

If the category is unitary and the direct-sum inclusions are isometries, this isomorphism should be unitary. Status: pending. The finite configuration-space coordinate theorem above proves only the basis-dependent vector-space skeleton of this target after sector bases have been supplied.

Theorem 4.9 (Finite Fibonacci braid matrix relation)**.** *Let*

$$B_{12} = \begin{pmatrix} q^2 & 0 \\ 0 & -q \end{pmatrix}, \qquad B_{23} = c \begin{pmatrix} -1 & qs \\ qs & q^2 \end{pmatrix}.$$

If

$$c(q+1) = q, \qquad qs^2 = q^2 + q + 1,$$

then

$$B_{12}B_{23}B_{12} = B_{23}B_{12}B_{23}.$$

This is the algebraic core of the sourced Fibonacci relation with

$$q = \exp(-2\pi i/5), \qquad \phi = (1+\sqrt{5})/2, \qquad s = \sqrt{\phi}, \qquad c = q/(q+1),$$

where the source writes $B_{23} = q(q+1)^{-1}\begin{pmatrix} -1 & q\sqrt{\phi} \\ q\sqrt{\phi} & q^2 \end{pmatrix}$ *and* $B_{12}B_{23}B_{12} - B_{23}B_{12}B_{23} = 0$. *Lean proves the parameterised algebra in* `Formalisation/Fibonacci/BraidMatrices.lean`*; Julia checks the specialisation numerically at high precision; WolframScript checks it exactly. AF node* `1.3.10` *has been accepted by verifier* `verifier-fib-braid-001`*. This result is not a categorical braid group representation theorem.*

Theorem 4.10 (Finite braid-matrix unitarity by conjugation)**.** *Let I be a finite index set and let $F, R \in M_I(\mathbb{C})$. Assume*

$$F^\dagger F = I, \qquad FF^\dagger = I, \qquad R^\dagger R = I, \qquad RR^\dagger = I.$$

Define

$$B = FRF^\dagger.$$

Then

$$B^\dagger B = I, \qquad BB^\dagger = I.$$

This is the finite matrix algebra behind the locally sourced statements that fusion-tree changes are represented by unitary F-matrices, braidings give unitary R-matrices that are diagonal in a suitable basis, and a braid matrix is

$$B = FRF^{-1}.$$

For the sourced five-dimensional Fibonacci adjacent-label basis, the matrix F has three 1's and the nontrivial Fibonacci block

$$\begin{pmatrix} \phi^{-1} & \phi^{-1/2} \\ \phi^{-1/2} & -\phi^{-1} \end{pmatrix},$$

and

$$R = \operatorname{diag}\big(e^{4\pi i/5}, e^{-3\pi i/5}, e^{-3\pi i/5}, e^{4\pi i/5}, e^{-3\pi i/5}\big).$$

Lean proves the generic finite-matrix theorem in `Formalisation/LinearAlgebra/Isometry.lean`*. Julia checks this five-dimensional sourced specialisation numerically at high precision, and WolframScript checks it exactly. AF node* `1.3.11` *has been accepted by verifier* `verifier-fib-braid-unitary-001`*. The extracted text around one displayed formula is OCR-ambiguous, so the local PDF page and the later extracted line giving $B = FRF^{-1}$ were both used. This theorem does not construct categorical braidings, conformal blocks, mapping-class-group representations, or analytic monodromy.*

Theorem 4.11 (Braid group action target)**.** *If $\mathcal{C}$ is braided, the neighbouring exchange maps induced by*

$$c_{Q,Q} : Q \otimes Q \to Q \otimes Q$$

should satisfy

$$B_i B_{i+1} B_i = B_{i+1} B_i B_{i+1}, \qquad B_i B_j = B_j B_i \quad |i-j| \geq 2.$$

Status: pending.

Definition 4.12 (Fine-graining isometry)**.** *For $M > N$, a fine-graining map is a morphism or finite matrix*

$$E_{N\to M} : Q^{\otimes N} \to Q^{\otimes M}$$

such that

$$E^{\dagger}_{N\to M} E_{N\to M} = \mathrm{id}_{Q^{\otimes N}} .$$

For a charge sector W, the induced map is

$$V^{W}_{N\to M}(f) = E_{N\to M} \circ f.$$

In finite coordinate form, after bases have been chosen, this definition is formalised as a structure consisting of a matrix E and the equation

$$E^{\dagger} E = I.$$

The Lean structure is `FiniteFineGraining` in `Formalisation/LinearAlgebra/FineGraining.lean`. It packages the validated finite consequences: ascending unitality, postcomposition Gram preservation, state-embedding trace preservation, expectation preservation, ascending adjoint preservation

$$E^{\dagger} O^{\dagger} E = (E^{\dagger} O E)^{\dagger},$$

logical-lift retraction, logical lift of the identity

$$(EI)E^{\dagger} = EE^{\dagger},$$

and the code projection identities for $EE^{\dagger}$. AF node `1.4.12` has been accepted by verifier `verifier-finite-fine-gr` AF node `1.4.14` has been accepted by verifier `verifier-finite-fg-star-lift-001` for the packaged adjoint-preservation and logical-identity consequences. This is only the finite coordinate definition; it does not construct $Q^{\otimes N}$, categorical morphisms, Hom-sector semantics, categorical channel semantics, or physical fine-graining maps in a fusion category.

Theorem 4.13 (Postcomposition isometry target)**.** *If $E^{\dagger}E = \mathrm{id}$, then $V^{W}(f) = E \circ f$ is an isometry for every W. Conversely, if all simple-sector maps V^{W} are isometries and the Hom functor separates morphisms, then $E^{\dagger}E = \mathrm{id}$. Status: the finite-matrix forward direction is proved in `Formalisation/LinearAlgebra/Postcompose.lean`: if $E^{\dagger}E = I$, then*

$$(EF)^{\dagger}(EG) = F^{\dagger}G.$$

AF node `1.4.4` is accepted for this coordinate matrix theorem. The finite square-test converse is also proved: if

$$(EF)^{\dagger}(EG) = F^{\dagger}G$$

for every pair of square coarse-coordinate matrices F, G, then

$$E^{\dagger} E = I.$$

Indeed, take $F = I$ and $G = I$. AF node `1.4.8` is accepted for this finite matrix converse. The full categorical converse using semisimplicity and a separating Hom functor, and the categorical Hom-sector semantics, remain pending.

Theorem 4.14 (Component orthogonality target)**.** *For decompositions*

$$Q^{\otimes N} \cong \bigoplus_{\vec{a}} A_{\vec{a}}, \qquad Q^{\otimes M} \cong \bigoplus_{\vec{b}} B_{\vec{b}},$$

define

$$E_{\vec{b},\vec{a}} := \pi_{\vec{b}} E \iota_{\vec{a}}.$$

Then E is an isometry exactly when

$$\sum_{\vec{b}} E^{\dagger}_{\vec{b},\vec{a}} E_{\vec{b},\vec{a}'} = \delta_{\vec{a},\vec{a}'} \operatorname{id}_{A_{\vec{a}}} .$$

Status: the scalar finite-matrix version is proved in `Formalisation/LinearAlgebra/Component.lean`*: for a matrix E,*

$$E^{\dagger}E = I \quad \Longleftrightarrow \quad \sum_{b} \overline{E_{b,a}} E_{b,a'} = \delta_{a,a'}.$$

AF node `1.4.2` *is accepted only for this coordinate matrix theorem. Block Hom spaces and categorical direct-sum semantics remain pending.*

Theorem 4.15 (Block tensor isometry target)**.** *Let $M > N$, let the fine sites be partitioned into ordered blocks of sizes $r_i \geq 1$, and let*

$$\eta_i : Q \to Q^{\otimes r_i}$$

be isometries. Then

$$E^{\mathrm{block}}_{N \to M} = \eta_1 \otimes \cdots \otimes \eta_N$$

is an isometry. Status: the two-factor finite-matrix version is proved in `Formalisation/LinearAlgebra/Tensor.le` *if $E_1^{\dagger}E_1 = I$ and $E_2^{\dagger}E_2 = I$, then*

$$(E_1 \otimes E_2)^{\dagger}(E_1 \otimes E_2) = I.$$

AF node `1.4.3` *is accepted only for this Kronecker product theorem. The same Lean file also proves the heterogeneous three-factor finite matrix case: if $E_1^{\dagger}E_1 = I$, $E_2^{\dagger}E_2 = I$, and $E_3^{\dagger}E_3 = I$, then*

$$((E_1 \otimes E_2) \otimes E_3)^{\dagger}((E_1 \otimes E_2) \otimes E_3) = I.$$

AF node `1.4.10` *has been accepted by verifier* `verifier-tensor-three-het-001`*. The fixed-local-map induction is proved in* `Formalisation/LinearAlgebra/TensorPower.lean`*: if $E^{\dagger}E = I$, then every finite Kronecker tensor power $E^{\otimes N}$ is an isometry. Julia and WolframScript check exact rational examples through $N = 6$. AF node* `1.4.7` *has been accepted by verifier* `verifier-tensor-power-001`*. The arbitrary heterogeneous finite-coordinate induction is proved in* `Formalisation/LinearAlgebra/Heterogeneous` *For any natural number N, finite dependent families of fine and coarse coordinate index types, and local matrices E_i with $E_i^{\dagger}E_i = I$, the recursively ordered heterogeneous Kronecker product satisfies*

$$\big(E_0 \otimes E_1 \otimes \cdots \otimes E_{N-1}\big)^{\dagger}\big(E_0 \otimes E_1 \otimes \cdots \otimes E_{N-1}\big) = I.$$

AF node `1.4.16` *has been accepted by verifier* `verifier-het-tensor-n-001`*. Categorical tensor products, canonical associators, ordered partition semantics, and construction of local maps from fusion-category data remain pending.*

Theorem 4.16 (Unitary dressing target)**.** *If $U_M : Q^{\otimes M} \to Q^{\otimes M}$ is unitary and each η_i is isometric, then*

$$E_{N\to M} = U_M(\eta_1 \otimes \cdots \otimes \eta_N)$$

is isometric. Status: the finite-matrix calculation $(UE)^\dagger(UE) = I$ from $U^\dagger U = I$ and $E^\dagger E = I$ is proved in `Formalisation/LinearAlgebra/Isometry.lean`. The node has been accepted by verifier `verifier-mat-dress-001` only for this finite-matrix calculation. For three heterogeneous factors, the Lean file `Formalisation/LinearAlgebra/Tensor.lean` also proves that if U is unitary on the three-factor fine product, then

$$\big(U((E_1 \otimes E_2) \otimes E_3)\big)^\dagger U((E_1 \otimes E_2) \otimes E_3) = I.$$

AF node `1.4.11` has been accepted by verifier `verifier-dressed-three-tensor-001`. For fixed-local-map tensor powers, the Lean file `Formalisation/LinearAlgebra/TensorPower.lean` proves that if $E^\dagger E = I$, $E^{\otimes N}$ is the recursively defined Kronecker tensor power, and U is unitary on the fine tensor-power coordinate space, then

$$\big(UE^{\otimes N}\big)^\dagger\big(UE^{\otimes N}\big) = I.$$

Julia and WolframScript check exact rational rectangular examples with a nontrivial reversal permutation unitary through $N = 4$. AF node `1.4.15` has been accepted by verifier `verifier-dressed-tensor-power-001` The heterogeneous finite-coordinate theorem in `Formalisation/LinearAlgebra/HeterogeneousTensor.lean` also proves the arbitrary N-factor dressed case over the ordered recursive tensor index: if U is unitary on the fine tensor-product coordinate space, then

$$\big(U(E_0 \otimes \cdots \otimes E_{N-1})\big)^\dagger U(E_0 \otimes \cdots \otimes E_{N-1}) = I.$$

This is part of AF node `1.4.16`. Full categorical unitary dressing, ordered partition semantics, and physical construction of U remain pending.

5 Finite-Dimensional Linear Algebra

Theorem 5.1 (Polar section target)**.** *Let $B : H_{\text{fine}} \to H_{\text{coarse}}$ be a full-rank linear map between finite-dimensional Hilbert spaces. Define*

$$E := B^\dagger(BB^\dagger)^{-1/2}.$$

Then $E^\dagger E = \mathrm{id}_{H_{\text{coarse}}}$. Status: the conditional finite-matrix algebra is proved in `Formalisation/LinearAlgebra/Po` If a matrix R satisfies

$$R^\dagger(BB^\dagger)R = I,$$

then $E = B^\dagger R$ satisfies $E^\dagger E = I$. AF node `1.4.5` is accepted only for this conditional statement; existence and positivity of $(BB^\dagger)^{-1/2}$ remain pending.

There is also an accepted diagonal special case. Suppose $BB^\dagger$ is diagonal in the chosen coarse basis, with diagonal entries $g_\mu > 0$. Define

$$R_{\mu\mu'} = \begin{cases} (\sqrt{g_\mu})^{-1}, & \mu = \mu', \\ 0, & \mu \neq \mu'. \end{cases}$$

Then $R^\dagger = R$,

$$R^\dagger(BB^\dagger)R = I,$$

and $B^\dagger R$ is an isometry. Lean defines this matrix as `diagonalInvSqrt` and proves the inverse-square-root condition and isometry in `Formalisation/LinearAlgebra/Polar.lean`. Julia checks the diagonal case $g = (4, 9)$, and WolframScript checks the same case exactly. AF node `1.4.13` has been accepted by verifier `verifier-polar-diag-sqrt-001`. This does not prove the general positive matrix square-root theorem, full polar decomposition existence, the full-rank criterion, or categorical Hilbert-space semantics.

Theorem 5.2 (Amplitude formula target)**.** *For orthonormal bases, write*

$$B|\nu\rangle_{\text{fine}} = \sum_{\mu} \beta_{\mu\nu} |\mu\rangle_{\text{coarse}}$$

and

$$G_{\mu\mu'} = \sum_{\nu} \beta_{\mu\nu} \overline{\beta_{\mu'\nu}}.$$

Then $E|\mu\rangle = \sum_\nu A_{\nu\mu}|\nu\rangle$ with

$$A_{\nu\mu} = \sum_{\mu'} \overline{\beta_{\mu'\nu}}\, (G^{-1/2})_{\mu'\mu}.$$

If μ does not mix with other coarse labels, then

$$A^{\mu}_{\nu} = \frac{\overline{\beta_{\mu\nu}}}{\sqrt{\sum_{\nu'} |\beta_{\mu\nu'}|^2}}.$$

Status: `Formalisation/LinearAlgebra/Polar.lean` proves the coordinate entry formula for $E = B^\dagger R$, with R standing for the inverse-square-root factor. The finite one-row no-mixing corollary is proved in `Formalisation/LinearAlgebra/NoMixing.lean`: for a blocking row β_ν, the matrix $B^\dagger R$ has entries

$$\frac{\overline{\beta_{\nu}}}{\sqrt{\sum_{\nu'} |\beta_{\nu'}|^2}}$$

and is an isometry when the denominator is nonzero. AF node `1.4.9` is accepted for exactly this finite matrix corollary. Full square-root and polar decomposition semantics remain pending.

Theorem 5.3 (Ascending channel target)**.** *For an isometry V, the observable map*

$$\mathsf{A}(O) = V^\dagger O V$$

is unital, completely positive, star-preserving, and expectation-value preserving against embedded states. The finite matrix unital, star-preserving, and retraction identities are formalised in `Formalisation/LinearA` The finite matrix trace identities

$$\operatorname{tr}(E\rho E^\dagger) = \operatorname{tr}(\rho) \quad \textit{when } E^\dagger E = \mathbf{1}$$

and

$$\operatorname{tr}\big((E\rho E^\dagger)O\big) = \operatorname{tr}\big(\rho(E^\dagger O E)\big)$$

are formalised in `Formalisation/LinearAlgebra/Trace.lean`. The same file proves the square-form identity

$$K^\dagger(X^\dagger X)K = (XK)^\dagger(XK),$$

which is a finite-dimensional positivity witness. More generally, call a finite matrix O Gram-form positive if there is a finite witness index set and a matrix X, possibly rectangular, with

$$O = X^\dagger X.$$

If O is a fine observable with such a witness, then

$$E^\dagger O E = (XE)^\dagger (XE),$$

so the ascending congruence preserves this Gram-form witness. In Mathlib's finite quadratic-form definition of positive semidefinite matrices, the same file proves the order-theoretic implication

$$O \geq 0 \implies E^\dagger O E \geq 0.$$

This is the ordinary finite matrix fact that a positive semidefinite matrix remains positive semidefinite after congruence by E. For every finite auxiliary coordinate set R, the same file also proves the amplified finite statement

$$O \geq 0 \implies (I_R \otimes E)^\dagger O (I_R \otimes E) \geq 0$$

for matrices O on $R \times$ fine. This proves amplified positivity for the one-Kraus congruence $I_R \otimes E$. If $\{K_i\}_{i \in S}$ is any finite family of matrices with common source and target, then the finite congruence sums also preserve positivity:

$$O \geq 0 \implies \sum_{i \in S} K_i^\dagger O K_i \geq 0$$

and, for every finite auxiliary coordinate set R,

$$O \geq 0 \implies \sum_{i \in S} (I_R \otimes K_i)^\dagger O (I_R \otimes K_i) \geq 0.$$

For a finite family of matrices $\{K_i\}_{i \in S}$, define

$$\mathsf{A}(O) = \sum_{i \in S} K_i^\dagger O K_i, \qquad \mathsf{S}(\rho) = \sum_{i \in S} K_i \rho K_i^\dagger.$$

If

$$\sum_{i \in S} K_i^\dagger K_i = I,$$

then

$$\mathsf{A}(I) = I, \qquad \mathrm{tr}(\mathsf{S}(\rho)) = \mathrm{tr}(\rho).$$

For every finite family, the same finite matrix formalisation proves

$$\mathsf{A}(O^\dagger) = \mathsf{A}(O)^\dagger$$

and

$$\mathrm{tr}(\mathsf{S}(\rho) O) = \mathrm{tr}(\rho\, \mathsf{A}(O)).$$

This is finite-dimensional matrix algebra. It does not assert that a particular physical refining channel has a sourced Kraus representation. For the logical lift

$$\mathsf{L}(O) = V O V^\dagger,$$

the finite matrix formalisation also proves

$$\mathsf{L}(I) = V V^\dagger, \qquad (V V^\dagger)^2 = V V^\dagger, \qquad (V V^\dagger)^\dagger = V V^\dagger.$$

Thus the lifted coarse identity is the code-subspace projection in the finite matrix sense, not the full fine identity. Status: AF nodes `1.5.1`, `1.5.2`, `1.5.8`, `1.5.9`, `1.5.11`, and `1.5.12`, `1.5.13`, and `1.5.14` are accepted for these finite-matrix identities only. Full categorical channel semantics, categorical subobject semantics, full spectral projection theory, and a sourced physical Kraus-representation theorem remain pending.

6 Fibonacci Data

Definition 6.1 (Fibonacci labels and fusion rules)**.** *The Fibonacci model has two simple labels,* $\mathbf{1}$ *and* τ*, with*

$$\mathbf{1} \otimes \mathbf{1} = \mathbf{1}, \quad \mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \mathbf{1} = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau.$$

Local source: `references/text/FibonacciAnyonModels.txt:128`*,* `:129`*, and* `:133`*. Status: formalised in* `Formalisation/Fibonacci/FusionRules.lean`*. AF node* `1.3.8` *has been accepted by verifier* `verifier-fib-fusion-001` *only for the finite multiplicity table. The same file instantiates the finite skeletal fusion-data structure from AF node* `1.2.7` *and proves the multiplicity-free condition and unit total multiplicities for this Fibonacci table.*

Definition 6.2 (Golden ratio and total quantum dimension)**.** *Define*

$$\varphi := \frac{1+\sqrt{5}}{2}, \qquad d_{\mathbf{1}} = 1, \qquad d_\tau = \varphi.$$

Define

$$D^2 := 1 + \varphi^2.$$

Local sources: `references/text/FibonacciAnyonModels.txt:216`*,* `:218`*,* `:304`*, and* `references/text/TrebstSho`
`:325`*.*

Theorem 6.3 (Fibonacci constant identities)**.** *The following identities hold:*

$$\varphi^2 = \varphi + 1, \qquad \varphi^{-1} = \varphi - 1, \qquad \varphi^{-2} + \varphi^{-1} = 1, \qquad D^2 = 2 + \varphi.$$

Status: Adversarial Proof Framework node `1.3.1` *submitted for verification and accepted by verifier* `verifier-fib-alg-001`*. Lean, Julia, and WolframScript checks pass. The accepted scope is the real-number algebra claim about* φ *and* D^2*, not a broader category-theoretic Fibonacci theorem.*

Proof. The Lean proof is in `Formalisation/Fibonacci/Basic.lean`. It unfolds φ, uses $(\sqrt{5})^2 = 5$, and solves the real algebra. Julia checks the equations numerically at 256-bit precision. WolframScript checks the same equations by exact symbolic simplification. □

Definition 6.4 (Fibonacci F-matrix)**.** *The nontrivial* F*-matrix used initially is*

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}.$$

Local sources: `references/text/FibonacciAnyonModels.txt:272`*,* `:297`*,* `:301`*,* `:304`*, and* `references/text/Gold`

Theorem 6.5 (Fibonacci F-matrix exact identities)**.** *The matrix above satisfies*

$$F^\dagger F = I, \qquad F^2 = I.$$

Status: ready for adversarial verification in node `1.3.2`*; Lean, Julia, and WolframScript checks pass for the parameterised theorem* $\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ *under* $a^2 + b^2 = 1$*. The node has been accepted by verifier* `verifier-fib-mat-001` *as a real matrix algebra claim. The categorical meaning of the* F*-move is not included in that acceptance.*

Definition 6.6 (General binary Fibonacci map)**.** *Let $Q = \mathbf{1} \oplus \tau$. A binary map*

$$\eta : Q \to Q \otimes Q$$

has coefficients

$$\eta_{\mathbf{1}} = x v^{\mathbf{1}}_{\mathbf{1}\mathbf{1}} + y v^{\mathbf{1}}_{\tau\tau}$$

and

$$\eta_\tau = p v^\tau_{\tau\mathbf{1}} + q v^\tau_{\mathbf{1}\tau} + r v^\tau_{\tau\tau}.$$

Theorem 6.7 (Binary Fibonacci isometry target)**.** *The map η is an isometry exactly when*

$$|x|^2 + |y|^2 = 1, \qquad |p|^2 + |q|^2 + |r|^2 = 1.$$

There is no cross-condition between the $\mathbf{1}$ and τ sectors because $\mathrm{Hom}(\mathbf{1}, \tau) = 0$*. Status: ready for adversarial verification in node `1.3.6` for the coordinate matrix theorem after choosing the orthogonal binary fusion-tree basis. The Lean proof is in `Formalisation/Fibonacci/Binary.lean`. The node has been accepted by verifier `verifier-fib-binary-001` only at this coordinate-matrix level.*

Definition 6.8 (Perron–Frobenius binary amplitudes)**.** *For a finite fusion channel $a \to b \otimes c$, the project handoff proposes the topological one-step coefficient*

$$A^a_{bc} = \sqrt{\frac{d_b d_c}{d_a D^2}},$$

where d_a is the positive Frobenius–Perron dimension and

$$D^2 = \sum_x d_x^2.$$

For Fibonacci labels $d_{\mathbf{1}} = 1$, $d_\tau = \varphi$, and $D^2 = 1 + \varphi^2$. On the five allowed binary channels this gives

$$\eta^{\mathrm{PF}}_{\mathbf{1}} = \frac{1}{D} v^{\mathbf{1}}_{\mathbf{1}\mathbf{1}} + \frac{\varphi}{D} v^{\mathbf{1}}_{\tau\tau}$$

and

$$\eta^{\mathrm{PF}}_\tau = \frac{1}{D} v^\tau_{\tau\mathbf{1}} + \frac{1}{D} v^\tau_{\mathbf{1}\tau} + \frac{\sqrt{\varphi}}{D} v^\tau_{\tau\tau}.$$

Theorem 6.9 (Perron–Frobenius normalisation)**.** *In the finite skeletal Fibonacci model, the dimension vector*

$$d_{\mathbf{1}} = 1, \qquad d_\tau = \varphi$$

is an eigenvector of the τ-fusion matrix with eigenvalue φ. The five coefficient equalities from $A^a_{bc} = \sqrt{d_b d_c / (d_a D^2)}$ are

$$A^{\mathbf{1}}_{\mathbf{1}\mathbf{1}} = \frac{1}{D}, \quad A^{\mathbf{1}}_{\tau\tau} = \frac{\varphi}{D}, \quad A^\tau_{\tau\mathbf{1}} = \frac{1}{D}, \quad A^\tau_{\mathbf{1}\tau} = \frac{1}{D}, \quad A^\tau_{\tau\tau} = \frac{\sqrt{\varphi}}{D}.$$

The scalar normalisations are

$$\frac{1}{D^2} + \frac{\varphi^2}{D^2} = 1, \qquad \frac{1}{D^2} + \frac{1}{D^2} + \frac{\varphi}{D^2} = 1.$$

Combined with the binary coordinate isometry theorem, these identities prove that the 5×2 coordinate matrix for η^{PF} satisfies

$$(\eta^{\mathrm{PF}})^\dagger \eta^{\mathrm{PF}} = I.$$

The scalar identities are accepted in AF node `1.3.3`. The coordinate-level matrix isometry is accepted in AF node `1.3.13` by verifier `verifier-fib-pf-binary-001`*. The finite Fibonacci dimension-formula coefficient calculation is accepted in AF node `1.3.15` by verifier* `verifier-fib-pf-dim-amp-001`*. This does not prove the arbitrary fusion-category amplitude theorem, categorical Perron–Frobenius naturality, string-net fixed-point semantics, or construction of the fusion-tree basis.*

Definition 6.10 (Coassociative candidate)**.** *The positive real candidate is*

$$\eta_{\mathbf{1}}^{\mathrm{coassoc}} = \varphi^{-1} v_{\mathbf{11}}^{\mathbf{1}} + \varphi^{-1/2} v_{\tau\tau}^{\mathbf{1}}$$

and

$$\eta_{\tau}^{\mathrm{coassoc}} = \varphi^{-1} v_{\tau\mathbf{1}}^{\tau} + \varphi^{-1} v_{\mathbf{1}\tau}^{\tau} + \varphi^{-3/2} v_{\tau\tau}^{\tau}.$$

Status: scalar equations pass Julia and WolframScript; the full associator proof is pending. The scalar equations and displayed two-component F-comparison vector are proved in `Formalisation/Fibonacci/Coass` *this is ready for adversarial verification in node `1.3.4`. The node has been accepted by verifier* `verifier-coassoc-scalar-001` *only at this scalar level. It does not prove categorical coassociativity, uniqueness, or associator semantics.*

Theorem 6.11 (Positive scalar coassociative solution)**.** *Let x, y, r be positive real numbers. Suppose*

$$x^2 + y^2 = 1, \qquad 2x^2 + r^2 = 1,$$

and

$$r^2 = \varphi^{-3/2} xy.$$

Then

$$x = \varphi^{-1}, \qquad y = \varphi^{-1/2}, \qquad r = \varphi^{-3/2}.$$

In Lean the coefficient $\varphi^{-3/2}$ is represented as $\varphi^{-1}\varphi^{-1/2}$, and $r = \varphi^{-3/2}$ is represented by $r = \varphi^{-1}\varphi^{-1/2}$. Lean now proves the scalar identities

$$\varphi^{-1/2} = (\sqrt{\varphi})^{-1}, \qquad \varphi^{-3/2} = \varphi^{-1}(\sqrt{\varphi})^{-1},$$

and the squared positive-root form

$$r^2 = (\varphi^{-1})^3, \qquad r = \sqrt{(\varphi^{-1})^3}$$

for the positive candidate r. The proof is in `Formalisation/Fibonacci/Coassoc.lean`*; Julia checks the eliminated polynomial roots and fractional-power identities numerically at high precision; WolframScript checks the exact quantified uniqueness implication and fractional-power identities. AF node `1.3.12` has been accepted by verifier* `verifier-coassoc-unique-001`*; AF node `1.3.16` has been accepted by verifier* `verifier-coassoc-frac-pow-001`*. This remains only scalar algebra and does not prove categorical coassociativity, associator semantics, fusion-tree basis comparison, or existence of a categorical map.*

Theorem 6.12 (Coordinate isometry of the strict binary candidate)**.** *Use the fixed binary coordinate basis whose rows are the five allowed channels*

$$\mathbf{1} \to \mathbf{11}, \quad \mathbf{1} \to \tau\tau, \quad \tau \to \tau\mathbf{1}, \quad \tau \to \mathbf{1}\tau, \quad \tau \to \tau\tau,$$

and whose columns are the two input labels $\mathbf{1}, \tau$. The coefficient choice displayed in `handoff.md` *section 9.6 is the matrix*

$$\eta_{\text{coassoc}} = \begin{pmatrix} \varphi^{-1} & 0 \\ \varphi^{-1/2} & 0 \\ 0 & \varphi^{-1} \\ 0 & \varphi^{-1} \\ 0 & \varphi^{-3/2} \end{pmatrix}.$$

It satisfies

$$\eta_{\text{coassoc}}^{\dagger} \eta_{\text{coassoc}} = I_2.$$

Lean defines this matrix as `CoassocBinaryEta` *and proves* `CoassocBinaryEta.conjTranspose * CoassocBinaryEta = 1` *in* `Formalisation/Fibonacci/Coassoc.lean`*, using the binary coordinate isometry theorem in* `Formalisation/Fibonacci/Binary.lean` *and the scalar norm equalities above. Julia checks the four entries of the Gram matrix numerically at high precision, and WolframScript checks the exact symbolic identity. AF node* `1.3.17` *has been accepted by verifier* `verifier-coassoc-binary-iso-00` *This is only a fixed-basis coordinate isometry; it does not prove categorical coassociativity, associator semantics, fusion-tree basis comparison, or existence of a categorical map.*

7 Renormalization Input and Scaling Operators

Definition 7.1 (Conformal weight input)**.** *The local paper* `references/IsingLikeFibonacciAnyonsKZ.pdf` *identifies the Fibonacci anyon field τ with the spin-1 field in the level-3 $SU(2)$ Wess–Zumino–Witten model, and its table of primary-field conformal weights gives the spin-1 field the chiral conformal weight*

$$h_\tau = \frac{2}{5}.$$

For a diagonal theory in which the left and right chiral weights are both h_τ, the scaling dimension is their sum:

$$\Delta_\tau = h_\tau + \overline{h}_\tau = \frac{2}{5} + \frac{2}{5} = \frac{4}{5}.$$

This addition convention is sourced locally from `references/text/PoilblancOneDimensionalItinerantInteract` *The chiral exponent arithmetic used later for the Fibonacci renormalization coefficients is*

$$h_{\mathbf{1}} - 2h_\tau = 0 - 2 \cdot \frac{2}{5} = -\frac{4}{5}, \qquad h_\tau - 2h_\tau = \frac{2}{5} - 2 \cdot \frac{2}{5} = -\frac{2}{5}.$$

Squaring these powers in the normalisation denominators gives

$$2\left(h_{\mathbf{1}} - 2h_\tau\right) = -\frac{8}{5}, \qquad 2\left(h_\tau - 2h_\tau\right) = -\frac{4}{5}.$$

The exact rational equalities are proved in `Formalisation/Fibonacci/CFTWeights.lean` *and checked in Julia and WolframScript. AF node* `1.5.5` *has been accepted by verifier* `verifier-cft-weight-001`*. AF node* `1.5.7` *has been accepted by verifier* `verifier-cft-rg-exp-001` *for the exponent arithmetic. These formal results do not prove the Wess–Zumino–Witten construction, extraction of the source table, Wilsonian renormalization coefficients, physical operator product constants, or a physical renormalization eigenvalue.*

The handoff scaling-operator section also uses the rational exponent convention

$$\lambda_\tau = b^{-2/5}, \qquad \lambda_{\tau,n} = b^{-(2/5+n)}, \qquad \lambda_{\mathbf{1},n} = b^{-n}, \qquad \lambda_\tau^{\text{diag}} = b^{-4/5}.$$

Lean proves the corresponding rational exponents

$$-\frac{2}{5}, \qquad -\left(\frac{2}{5}+n\right), \qquad -n, \qquad -\frac{4}{5}$$

in `Formalisation/Fibonacci/CFTWeights.lean`*, and Julia and WolframScript check the same equalities through* $n = 6$*. AF node* `1.5.15` *has been accepted by verifier* `verifier-cft-desc-exp-001`*. This is only arithmetic for supplied weights and descendant labels; it does not construct descendant states, define real powers* b^{-w}*, prove diagonal renormalization, or construct physical scaling eigen-operators.*

Definition 7.2 (Blocking coefficients)**.** *For a scale factor* $b > 1$*, set* $\rho = b^{-1}$*. The local blocking coefficients are treated as input data:*

$$\beta^{\mathbf{1}}_{\mathbf{11}} = u_0, \quad \beta^{\mathbf{1}}_{\tau\tau} = u_I\rho^{-4/5},$$
$$\beta^{\tau}_{\tau\mathbf{1}} = j_L, \quad \beta^{\tau}_{\mathbf{1}\tau} = j_R, \quad \beta^{\tau}_{\tau\tau} = u_\tau\rho^{-2/5}.$$

The constants $u_0, u_I, j_L, j_R, u_\tau$ *are Wilsonian renormalization input data; they are not determined by scaling dimensions alone.*

Theorem 7.3 (Fibonacci renormalization amplitude target)**.** *Assuming the coefficients above, the no-mixing polar formula gives*

$$\eta^{\mathrm{RG}}_{\mathbf{1}}(\rho) = \frac{\overline{u_0}v^{\mathbf{1}}_{\mathbf{11}} + \overline{u_I}\rho^{-4/5}v^{\mathbf{1}}_{\tau\tau}}{\sqrt{|u_0|^2 + |u_I|^2\rho^{-8/5}}}$$

and

$$\eta^{\mathrm{RG}}_{\tau}(\rho) = \frac{\overline{j_L}v^{\tau}_{\tau\mathbf{1}} + \overline{j_R}v^{\tau}_{\mathbf{1}\tau} + \overline{u_\tau}\rho^{-2/5}v^{\tau}_{\tau\tau}}{\sqrt{|j_L|^2 + |j_R|^2 + |u_\tau|^2\rho^{-4/5}}}.$$

Status: computational checks in Julia and WolframScript pass for normalisation and for the finite one-row polar entry formula. The Lean file `Formalisation/Fibonacci/RGNoMixing.lean` *formalises the two rows as finite no-mixing vectors*

$$(u_0, u_I s_{\mathrm{vac}}), \qquad (j_L, j_R, u_\tau s_\tau),$$

where s_{vac} *and* s_τ *stand for the supplied powers* $\rho^{-4/5}$ *and* $\rho^{-2/5}$*. It proves the displayed denominator formulas, the normalized-conjugate entries, and row normalisation whenever the denominator is nonzero. It also proves that the squared norm of the* $\tau\tau$ *amplitude in the tau row is*

$$\frac{|u_\tau|^2 s_\tau^2}{|j_L|^2 + |j_R|^2 + |u_\tau|^2 s_\tau^2},$$

which becomes the handoff probability

$$P(\tau \to \tau\tau) = \frac{|u_\tau|^2\rho^{-4/5}}{|j_L|^2 + |j_R|^2 + |u_\tau|^2\rho^{-4/5}}$$

after the supplied substitution $s_\tau^2 = \rho^{-4/5}$*. Lean proves that this quantity is nonnegative and at most* 1 *when the denominator is positive. AF node* `1.3.14` *has been accepted by verifier* `verifier-fib-rg-nomix-rows-001`*. The scalar no-mixing normalisation theorem is accepted in AF node* `1.3.7`*; the one-row polar-section formula is accepted in AF node* `1.4.9`*; the probability formula is accepted in AF node* `1.5.16`*. The full polar-decomposition theorem, Wilsonian derivation of the renormalization coefficients, physical operator product constants, fractional-power analysis, plotting, physical probability derivation, and categorical basis semantics remain separate targets.*

Theorem 7.4 (Conditional diagonal scaling). *Let I be a finite set of scaling labels. For each $i \in I$, let e_i be the coordinate vector with value 1 at i and 0 elsewhere. If a linear map R is diagonal in this basis with entries λ_i, so that*

$$(Rx)_j = \lambda_j x_j,$$

then

$$Re_i = \lambda_i e_i.$$

In particular, if the entries are stipulated to be $\lambda_i = b^{-w_i}$, then e_i is an eigenvector with eigenvalue b^{-w_i}. The Lean file `Formalisation/LinearAlgebra/DiagonalScaling.lean` proves this statement, with Julia and WolframScript checks for the chiral pattern $w_i = h_i$ and the nonchiral pattern $w_i = \Delta_i$. Status: AF node `1.5.3` is accepted only for this conditional diagonal linear-algebra statement. The rational scaling exponents for the Fibonacci primary, descendant, and diagonal nonchiral examples are accepted separately in AF node `1.5.15`. These results do not prove real-power analysis, the diagonalisation of a physical renormalization group map, or the charge-only negative example.

Warning 7.5 (Charge-only data do not determine the full scaling spectrum). *The two-dimensional charge-only operator space $\mathrm{End}(Q)$ does not generally recover all conformal field theory scaling operators. To recover the full spectrum, labels must include data such as primary field, descendant level, and mixing sector.*

A concrete finite counterexample is as follows. In the span of two charge projectors, write

$$u = (1, 1), \qquad c = (1, -1).$$

For any complex number a, define the charge-only linear map that is diagonal in the basis $\{u, c\}$ by

$$R_a u = u, \qquad R_a c = a\, c.$$

In the charge-projector coordinates this map is

$$R_a = \begin{pmatrix} (1+a)/2 & (1-a)/2 \\ (1-a)/2 & (1+a)/2 \end{pmatrix}.$$

Thus the charge contrast eigenvalue is the chosen amplitude a. Taking $a = 1/2$ and $b = 32$ gives $b^{-2/5} = 1/4$, so the charge contrast eigenvalue is not forced to equal $b^{-2/5}$. The Lean file `Formalisation/LinearAlgebra/ChargeOnly.lean` proves this finite example, and Julia and WolframScript check it numerically and exactly. Status: AF node `1.5.4` is accepted only for this concrete finite counterexample.

8 Temperley–Lieb and Ising Secondary Target

Definition 8.1 (Ising-type fusion rules). *The secondary test category has simple labels $\mathbf{1}, \sigma, \epsilon$. The local physics source writes the third label as ψ; in this document ϵ means that same label. The fusion rules are*

$$\sigma \otimes \sigma = \mathbf{1} \oplus \epsilon, \quad \sigma \otimes \epsilon = \sigma, \quad \epsilon \otimes \epsilon = \mathbf{1}.$$

Together with the unit rule

$$\mathbf{1} \otimes a = a = a \otimes \mathbf{1},$$

these rules are multiplicity-free except that $\sigma \otimes \sigma$ has two simple summands. The local sources are `references/text/PenneysUnitaryFusionCategories.md:647`, `:649`, and `references/text/PoilblancOneDimen` through `:188`.

Theorem 8.2 (Finite Ising toy exponent and normalisation). *Use the nonchiral Ising dimensions*

$$\Delta_\sigma = \frac{1}{8}, \qquad \Delta_\epsilon = 1.$$

Then the exponent for the channel

$$\sigma \otimes \sigma \to \epsilon$$

is

$$\Delta_\epsilon - 2\Delta_\sigma = 1 - 2 \cdot \frac{1}{8} = \frac{3}{4}.$$

For a real parameter t, the squared numerator norm of the toy vector with coefficients

$$\frac{1}{\sqrt{2}}, \qquad \frac{1}{\sqrt{2}}, \qquad \frac{t}{2}$$

is

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{t}{2}\right)^2 = 1 + \frac{t^2}{4}.$$

Thus division by $\sqrt{1+t^2/4}$ scalar-normalises the toy vector whenever the source fusion-tree basis vectors are orthonormal. In the handoff application $t = \rho^{3/4}$, giving the denominator $\sqrt{1+\rho^{3/2}/4}$. The Lean proof is in `Formalisation/Ising/Basic.lean`; Julia and WolframScript check the finite table and rational arithmetic. AF node `1.8` has been accepted by verifier `verifier-ising-toy-001`. This theorem does not construct the full Ising category, a Temperley–Lieb quotient, Ising F-symbols, operator product expansions, a conformal-field-theory spectrum, or a physical renormalization map.

9 Continuum Structure and Conjectures

Definition 9.1 (Direct-limit continuum Hilbert space). *A coherent family of refinements satisfies*

$$E_{N\to N} = \mathrm{id}, \qquad E_{M\to L}E_{N\to M} = E_{N\to L}.$$

The continuum charge-sector Hilbert space is the completed direct limit

$$\mathcal{H}^W_\infty = \overline{\varinjlim_N \mathcal{H}^W_N}.$$

Status: the completed direct-limit Hilbert space remains pending. The finite coherence algebra for ascending observables and logical lifts is formalised below.

Theorem 9.2 (Coherent finite ascending channels and logical lifts). *Let I be a finite coordinate set. Suppose that, for $n \le m$, matrices*

$$E_{n,m} \in M_I(\mathbb{C})$$

satisfy

$$E_{n,n} = I, \qquad E_{m,l}E_{n,m} = E_{n,l} \quad (n \le m \le l),$$

and

$$E^\dagger_{n,m}E_{n,m} = I.$$

Define the finite ascending observable map by

$$A_{n,m}(O) = E^\dagger_{n,m} O E_{n,m}.$$

Define the finite logical-lift map by

$$L_{n,m}(O) = E_{n,m} O E_{n,m}^{\dagger}.$$

Then

$$A_{n,m}(I) = I$$

and, for $n \leq m \leq l$,

$$A_{n,m}(A_{m,l}(O)) = A_{n,l}(O).$$

The logical-lift maps satisfy

$$L_{n,n}(O) = O$$

and, for $n \leq m \leq l$,

$$L_{m,l}(L_{n,m}(O)) = L_{n,l}(O).$$

The Lean file `Formalisation/LinearAlgebra/CoherentSystem.lean` proves this finite matrix theorem. Julia and WolframScript check exact rational rotation examples. AF node `1.5.6` has been accepted by verifier `verifier-coherent-asc-001`, and AF node `1.5.10` has been accepted by verifier `verifier-coherent-lift-001`. This theorem does not construct a completed direct limit, a continuum Hilbert space, a continuum algebra, a categorical fine-graining system, or nonconstant growing Hilbert spaces.

Conjecture 9.3 (Coherent subdivision family). *A physically correct continuum fine-graining scheme should use a coherent family*

$$\eta^{(r)} : Q \to Q^{\otimes r}, \qquad r = 1, 2, 3, \ldots,$$

compatible with arbitrary ordered subdivisions.

Conjecture 9.4 (Topological fixed points). *For topological fixed points, the coherent family can be expressed using F-symbols, quantum dimensions, and local string-net identities.*

Conjecture 9.5 (Renormalization away from fixed points). *Away from topological fixed points, the coherent family is determined by the full Wilsonian renormalization trajectory, not by the fusion category alone.*

Conjecture 9.6 (Universal isometry). *Every physically admissible finite-lattice fine-graining map is, at the categorical level, an isometry*

$$E_{N \to M} : Q^{\otimes N} \hookrightarrow Q^{\otimes M}.$$

Conjecture 9.7 (Topology and weights). *The fusion category determines which fine-graining channels may be nonzero. Renormalization data determine the finite-scale coefficients. Isometry then determines the normalised amplitudes.*

Conjecture 9.8 (Braiding and order changes). *If a fine-graining interpolation allows particles to exchange spatial order, then the category must be braided, and the order-change components must use the braiding. Without braiding, fine-graining maps should preserve order or avoid crossings.*

10 Current Machine-Checked Artefacts

- Lean project: `lakefile.lean`, `lean-toolchain`, and `Formalisation/`.
- Lean build command: `lake build`.
- Julia check: `scripts/julia/fibonacci_checks.jl`.
- WolframScript check: `scripts/wolfram/fibonacci_exact.wls`.
- Command log: `results/CHECKS.md`.
- Master provenance ledger: `MASTER_PROVENANCE.md`.